DEPARTMENT OF ELECTRONIC & ELECTRICAL ENGINEERING

University of Strathclyde Engineering

EE472/EE972 Control Principles

Semester 2 Project Assignment 2024/2025

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Project sessions:

There will be three project sessions, two hours each, all in RC/446 and RC/448, Royal College Building

- 19th February, Wednesday, 9:00 am - 11:00 am (Week 5)
- 5th March, Wednesday, 9:00 am - 11:00 am (Week 7)
- 19th March, Wednesday, 9:00 am - 11:00 am (Week 9)

The implementation of Tasks 1-3 in Simulink/Matlab need to be checked by a designated TA by the end of the third lab session in Week 9. During the lab sessions, TAs will only check your implementation progress, NOT assess the quality of the work.

Project report submission

The project report should be submitted to MyPlace, as a single pdf document, maximum 20 A4 pages, by **5pm, Monday, 31st March 2025**. A late penalty of 10% per week will be applied after the submission deadline.

Project marking

- 5 marks for progress checking: Task 1 (2 marks), Task 2 (1 mark), Task 3 (2 marks)
- 20 marks for project report (Tasks 1-4)

Plagiarism

Please read the University's policy and guidance on Academic Integrity Guidance | University of Strathclyde

1. Project Description

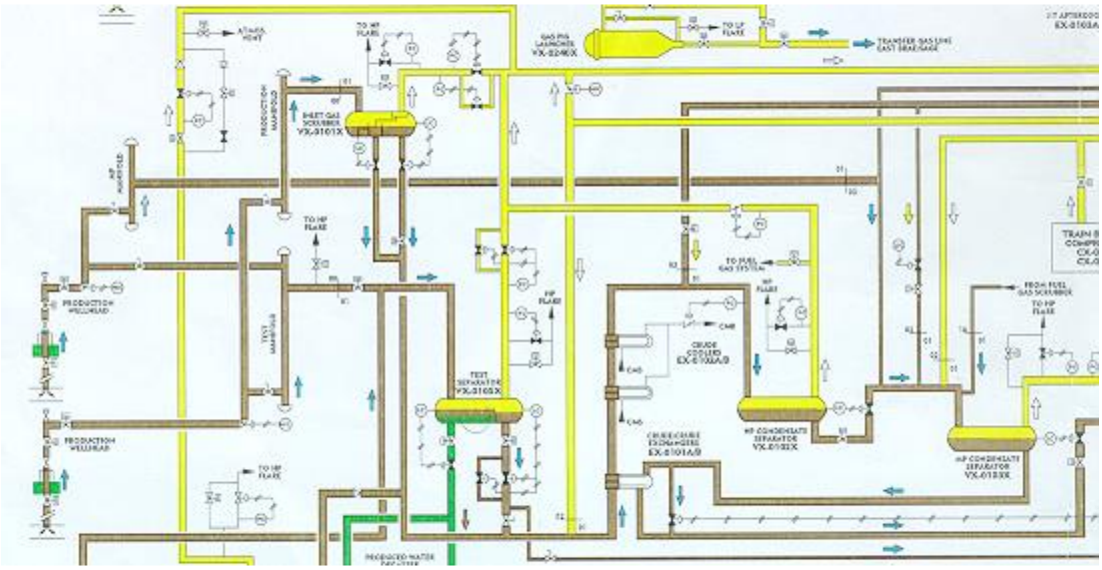

Figure 1. Slug catcher installation platform

Slug catcher is the part of the oil platform installation where the crude oil excavated from the bottom of the sea enters the platform (Fig.1). The crude oil consists of three main components: the oil, the gas and the water. Due to physical phenomena occurring during the transportation of crude oil through the long pipes, so called "slugs" are formed. These are high concentration of gas in the pipe travelling with the oil. When a slug of gas reaches the platform, the proportion of the three components in the crude oil changes. The purpose of the slug catcher installation is to act as a buffer, to remove the water, to separate the gas from the oil and to supply this 'cleaned' crude oil to other plants downstream in the installation.

For the purpose of this project we will consider a simplified scheme consisting of two tanks in the installation: the *Slug catcher vessel* and the *Free-water-knock-out vessel* (Fig. 2). In order to make the Slug catcher plant function effectively, the level of crude oil in both tanks has to be maintained between an upper and lower limit. The level is also used as surge capacity to ensure a continuous and constant flow of crude oil downstream to other units. Each of the tanks is equipped with a level control loop, which is a PI controller acting on a valve downstream. For the first tank the valve is a proportional one; for the second tank, it is an equal percentage valve. The process outputs to be controlled are tank levels h_1 and h_2, and the controller output are the valve lifts L_1 and L_2.

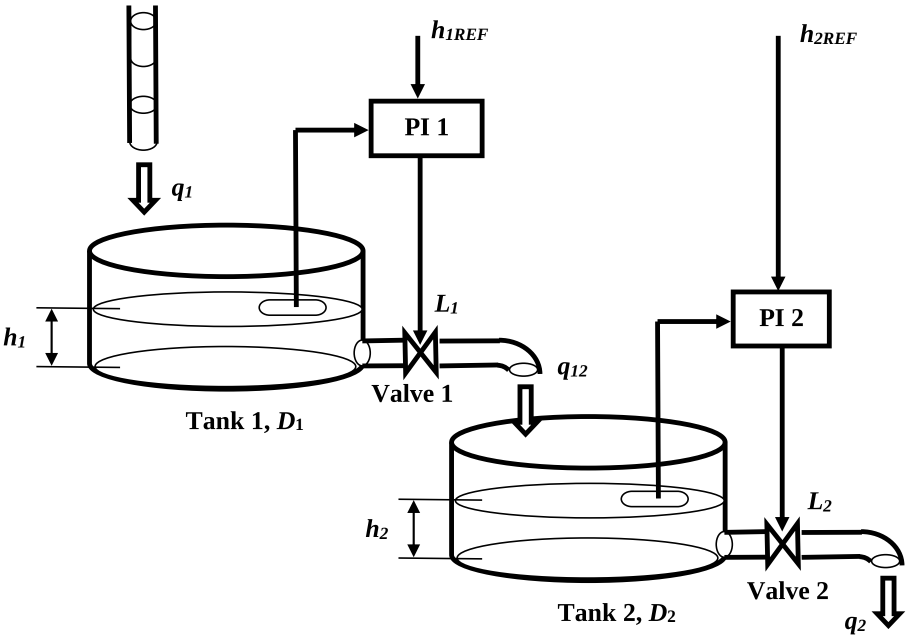

Figure 2. Schematic diagram of the two-tank system

The system consists of the following components:

Tanks

$$D_i \frac{d}{d\,t} h_i = q_{in} - q_{out}, \qquad i = 1,2$$

D_i - tank area (m^2), h_i- tank level (m), q_{in}- inlet flow rate (m^3/s), q_{out}- outlet flow rate (m^3/s).

Valve 1 (Proportional valve)

$$q_{12} = \left((1 - f_{1min})L_1 + f_{1min}\right)\sqrt{\frac{h_1}{h_{1min}} - 1} \qquad \text{with} \qquad f_{1min} = 0.02$$

Valve 2 (Equal percentage valve)

$$q_2 = 0.7R^{L_2 - 1}\sqrt{\frac{h_2}{h_{2min}} - 1} \qquad \text{with} \quad R = 25$$

L_1, L_2- valve lifts representing valve opening. They are the control signals u_1 and u_2.

Valve lift constraints

The valve lifts L_i are subject to the following two constraints on valve opening and varying velocity:

$$0 \le L_i \le 1, \qquad -0.1 \le \frac{d}{d\,t} L_i \le 0.1, \qquad i = 1,2$$

PI Controllers (controller output u_i are valve lifts L_i).

$$u_i(s) = \left(K_i + \frac{1}{T_i s}\right)\left(h_i(s) - h_{iREF}(s)\right), \qquad i = 1,2$$

Model Parameters

Table 1 Model parameters

Tank (i)	h_i^{min} (m)	h_i^{mid} (m)	h_i^{max} (m)	D_i (m^2)
1	0.4	1.0	1.6	1.324
2	0.2	0.7	1.2	2.02

Note: h_i^{min} and h_i^{max} are the minimum and maximum tank levels for the i-th tank, h_i^{mid} is the middle tank level. The units used in this model are in SI units, i.e., meters (m) and seconds (s)

External inflow

The flow to the first tank is a sum of a constant flow, a sinusoidal flow (slug formation) and a stochastic noise, which can be described as

$$q_1(t) = 0.5 + 0.1\, sin\left(\frac{2\pi}{50} t\right) + n(t),$$

where $n(t)$ is a band-limited white noise with noise power 0.02 and sampling time 10s.

Reference tank levels for Task 2

The reference heights for the two tanks are given in Table 2. When you design the PI controllers in Task 2, you should aim to let the tank levels follow these time-varying reference heights.

Table 2 Reference heights for two tanks

	Tank 1			**Tank 2**		
Time (s)	0-200	200-800	800-1,000	0-400	400-600	600-1,000
h_{iREF} (m)	0.8	0.96	0.8	0.5	0.45	0.5

2. Project Tasks

Task 1 Build up the system model in Simulink

(1a) Implement the two valves and plot their input-output characteristics (q vs. L) for minimum, maximum and half of tank level, h. **(10%)**

(1b) Implement the two tanks. **(5%)**

(1c) Implement the two PI controllers and the valve lift constraints. **(5%)**

(1d) Connect subsystems to build up the whole closed-loop system. **(10%)**

Task 2 PI controller design

(2a) Draw a block diagram for the system in Fig.2 representing the two subsystems and the interaction between them. It should include the tanks, controllers, valves and key signals. **(5%)**

(2b) Generate the external inflow q_1 and show its time profile in a figure. **(5%)**

(2c) Considering the reference signals for tank levels given in Table 2, tune the PI controllers to give satisfactory step responses. Justify your choice in PI controller design. Provide simulation results in figures. For each tank, you can put tank level and its reference signal in one figure to show the tracking performance, and the valve lifting in another figure to show the control activity. **(20%)**

Task 3 State feedback controller design

The above nonlinear system can be approximated by the following linear state-space model

$$\begin{bmatrix} \dot{h}_1 \\ \dot{h}_2 \end{bmatrix} = \begin{bmatrix} -0.47 & 0 \\ 0.31 & -0.41 \end{bmatrix} \begin{bmatrix} h_1 \\ h_2 \end{bmatrix} + \begin{bmatrix} -0.74 & 0 \\ 0.49 & -0.80 \end{bmatrix} \begin{bmatrix} L_1 \\ L_2 \end{bmatrix} \quad (1)$$

Design three state feedback controllers in (3a) and (3b) so that the desired tank levels, $h_{1REF} =$ 1.0 m, $h_{2REF} = 0.5$ m, can be achieved. In your simulation, set the initial states as $h_1(0) = 0.4$ m, $h_2(0) = 0.2$ m.

(3a) Design the state-feedback controller $\mathbf{u} = -\mathbf{K}\mathbf{x}$ to place the closed-loop poles at -1 and -2, and design the pre-compensator to remove tracking error at steady state. Change the desired poles to -0.2 and -0.5, design the feedback controller and pre-compensator again.

With these two controllers, show the step response results of control inputs and system outputs.

(10%)

(3b) Choose $\mathbf{Q} = \mathbf{R} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, design LQR to minimise

$$J = \frac{1}{2}\int_0^{\infty} [\boldsymbol{x}^T(t)\boldsymbol{Q}\boldsymbol{x}(t) + \boldsymbol{u}^T(t)\boldsymbol{R}\boldsymbol{u}(t)]dt$$

Also design the pre-compensator to remove tracking error at steady state. Apply the step input of h_{1REF} and h_{2REF}, show the time response of control inputs and system outputs.

(10%)

(3c) You have now developed three controllers, two by pole placement design and one by LQR design. Compare the three controllers in tracking performance and control activities. Use response figures or other quantitative data to support your discussion.

(10%)

Task 4 Discrete-time system analysis

Note: work on this task with manual solution, not computational tools such as Matlab/Simulink.

(4a) For the model in Equation (1), taking sampling time T, derive the discrete-time state space model using the Euler Forward method. Write the state equations and output equations in discrete-time state space form.

(5%)

(4b) For the model obtained in (4a), determine the range of sampling time T to assure that the discrete-time system is open-loop stable.

(5%)